Noise Traders in General Equilibrium

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Outline

1. Introduction

2. The Model

3. Results

"Noise Traders" in This Paper

Noise Traders are defined as agents with portfolio demand not equal to **FI-RE** demand.

(Full Information - Rational Expectations)

In a market with N risky assets,

- Let $\theta_{i,t} \in \mathbb{R}^N$ denote agent i's (dollar) portfolio demand at time t
- **FI-RE Demand** $\theta_{i,s}^{FI-RE} \in \mathbb{R}^N$ is the portfolio component of the solution to:

$$(C_{i,s}^{FI-RE}, \theta_{i,s}^{FI-RE})_{s \geq t} = \arg \max_{\{C_{i,s}, \theta_{i,s}\}_{s \geq t}} \mathbb{E}_t^{true} \left[U_i \left(\{C_{i,s}\}_{s \geq t} \right) \right],$$

  an EU problem under correct beliefs, standard wealth dynamics and correct return process
- **Noise Traders Demand** deviates from FI-RE demand: $\theta_{i,t} \neq \theta_{i,s}^{FI-RE}$

"Noise Traders" in Literature

- Noise Traders are *broadly* described as agents with trading behavior deviating from "benchmarks"

Category	Benchmark	Deviation Margin	Explanation
Informational	Full Information Rational Expectations (FI-RE)	Information Structure	Agents have distorted expectations about fundamentals.
Constraint-Based	Frictionless Optimization	Feasible Set	Rational agents with complete information, but portfolio demand is distorted by market frictions.
Preference-Based	Standard Expected Utility	Preferences	Agents are fully informed and optimizing, but preferences depart from EU
Reduced-Form Demand Shocks	Demand Fully Derived from Optimization	Derivation of Demand	Asset demand includes an exogenous component not derived from fundamentals nor utility maximization.

references

- In this paper, we focus on **informational distortions** relative to a FI-RE benchmark.

"Noise Traders" in General Equilibrium

Natural question to ask: "*Why* and *how* do agents deviate?"

- A substantial literature is devoted to studying these questions.
- **Not the focus of this paper.**

- Noise traders are *taken as given* in this model with the simplest form of demand distortion
- The focus is, instead, on the **general equilibrium consequences** of their behavior. Specifically:
  1. Deviations from the efficient-market benchmark
     - *How does the equilibrium pricing kernel depart from the FI-RE outcome?*
  2. Wealth distribution dynamics
     - *Do noise traders accumulate or lose wealth in equilibrium?*

Arbitrage Pricing Theory (APT)

- As a reference framework for general equilibrium pricing, we turn to **Arbitrage Pricing Theory:**
  1. Arbitrage-free markets imply the Euler equation $\boxed{\mathbb{E}_t\left[M_{t+dt}R_{k,t+dt}\right] = M_t}$

     (M_t: Stochastic Discount Factor (SDF), $R_{k,t}$: the return diffusion of asset k)
  2. If risk is driven by Brownian shocks $\mathbf{B}_t$, the SDF evolves as $\boxed{\frac{dM_t}{M_t} = -r_t\,dt - \lambda_t^\top d\mathbf{B}_t}$

     (r_t: risk free rate, λ_t : market prices of risk)
  3. Substituting this into the Euler equation yields $\boxed{\mathbb{E}_t\left[R_k - r\right] = \beta_k^\top \lambda}$

     where β_k is the collection of factor loadings on asset k.
- In other words, understanding equilibrium pricing boils down to **understanding the SDF** M_t, whose diffusion coefficients r_t and λ_t are functions of the underlying economic state.
- Relative to FI-RE, noise traders *expand* the state dependence.

Arbitrage Pricing with Noise Traders

- The state vector of a FI-RE economy is $\mathbf{x}_t$ = (fundamentals)
- With noise traders, the equilibrium state vector expands to

$$\mathbf{x}_t = (\text{fundamentals, behavioral states, wealth distribution})$$

  - Behavioral states represent shocks that distort noise traders' asset demands
  - Wealth distribution affects aggregate risk-bearing capacity in equilibrium
- The SDF M_t (due to r_t and λ_t) is a function of the equilibrium state vector

$$M_t = M(\mathbf{x}_t)$$

  - So noise traders do not change the arbitrage structure of the economy; they enlarge the state space that governs the pricing kernel and risk premia

What This Paper Does

- **Model Setup**
  - Continuous-time general equilibrium with perpetual youth
    in simple endowment (Lucas tree) & production (AK) environments
  - Two types of agents (rational & noise), belief distortions captured by a behavioral state
- **Equilibrium Characterization**
  - Derive optimal policies, solve for equilibrium prices and wealth distribution
  - Characterize the equilibrium pricing kernel
- **Quantitative Implementation**
  - Numerical solver for coupled HJB – market clearing system
  - Compute equilibrium objects over the state space, simulate economy

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Production

1. **Endowment Economy** (Lucas, 1978)

$$Y(t) = D(t), \quad D(t) = \bar{D}e^{z(t)}, \quad dz(t) = \mu_z dt + \sigma_z dB_f(t)$$

   - Output is an exogenous dividend stream $D(t)$

2. **Production Economy** (Brunnermeier and Sannikov, 2014)

$$Y(t) = \bar{A}K(t), \quad \frac{dK(t)}{K(t)} = (\iota(t) - \delta)\, dt + \sigma_K dB_f(t)$$

   - Productivity $\bar{A}$ is constant; $K(t)$ evolves endogenously through investment $\iota(t)$.

- $dB_f(t)$ – a single, *fundamental* Brownian shock
- Fundamental shocks are *i.i.d.* in both economies

Financial Assets

(c.f.) $X(t)$ *is the* ***aggregate state vector*** *with* $dX(t) = \mu_X(X,t)dt + \Sigma_X(X,t)^\top dB_X(t)$

Investors have access to two financial assets:

1. Locally **Risk-free bond** $b(t)$ in zero net supply

$$\frac{db}{b} = r(X,t)dt$$

2. **Risky Asset** (claim on the productive asset) with instantaneous return:

$$dR \equiv \frac{dP + d\Pi}{P} = \mu_R(X,t)dt + \Sigma_R(X,t)^\top dB_X$$

- P, μ_R, Σ_R are all endogenous. The payout $d\Pi$ satisfies

$$d\Pi = \begin{cases} D\,dt & \text{in the endowment economy} \\ \left[\bar{A} - \iota - \frac{\phi_I}{2}(\iota - \delta)^2\right] K\,dt & \text{in the production economy} \end{cases}$$

Demographics: Perpetual Youth (Yaari, 1965; Blanchard, 1985)

- **Perpetual Youth.** Continuous time, constant unit mass population
  - Poisson exit rate λ; entry flow λ
- **Annuity Market.**
  - No operative bequests; wealth redistributed across the living via actuarially fair annuities
  - Effective return $r + \lambda$ on wealth exactly offsets mortality-adjusted discount $\rho + \lambda$
- **Government Transfers.** fiscal transfer mechanism preventing permanent zero-wealth
  - $dT(t) = \tau Y(t)dt \implies W_{birth} = \frac{\tau Y(t)}{\lambda}$

Heterogeneous Agents

- 2 types of agents: **Noise Traders** $\mathcal{N}$ and **Rational Traders** $\mathcal{R}$:
  - Fixed fraction $\alpha \in [0, 1]$ are noise traders, the remaining $1 - \alpha$ are rational
- Type $\mathcal{R}$ and Type $\mathcal{N}$ differ only in the probability measure used to evaluate asset payoffs
  - Rational agents use the **objective** measure $\mathbb{P}$
  - Noise Trader agents use a **distorted** subjective measure $\tilde{\mathbb{P}}^{(i)} \sim \mathbb{P}$
- Belief distortion is driven by a "Subjective Wedge" $s(t)$
  - $s(t)$ is a non-fundamental wedge in subjective beliefs about expected returns

    DeLong et al. (1990, 1991), Barberis, Shleifer & Vishny (1998), Baker & Wugler (2006, 2007), Barberis (2018)
  - $s(t)$ enters directly into $\tilde{\mu}_R^{(i)}$ (subjective drift); preferences, endowments are identical across types

Individual Subjective Expected Return

- Type $\mathcal{N}$(noise) agents evaluate asset payoffs under *subjective* measure $\tilde{\mathbb{P}}^{(i)}$
  - $\tilde{\mu}_R^{(i)} \equiv \frac{1}{dt}\mathbb{E}_t^{\tilde{\mathbb{P}}^{(i)}}[dR]$ is the **Subjective Expected Return** for type $\mathcal{N}$ investor i
- $\mathbb{P} \sim \tilde{\mathbb{P}}^{(i)}$ over finite horizons (Girsanov): perfect agreement on instantaneous volatility Σ_R
  - Distortion is strictly confined to the drift: $d\tilde{B}_X^{(i)} = dB_X - \left(\frac{\tilde{\mu}_R^{(i)} - \mu_R}{\Sigma_R}\right) dt$
- $\tilde{\mu}_R^{(i)}$ evolves according to the stochastic differential equation:

$$d\tilde{\mu}_R^{(i)} = ds + \kappa\left(dR - \tilde{\mu}_R^{(i)} dt\right)$$

where ds is the prevailing subjective wedge and $\kappa \geq 0$ is a constant-gain learning rate

  - (Initial condition) $\tilde{\mu}_R^{(i)}(u) = \mu_R(u) + s(u)$, at i's time of entry u

Aggregate Subjective Expected Return

- Let $\tilde{\mu}_R^{\mathcal{N}}$ be the Aggregate (demographic-weighted) Subjective Expected Return for type $\mathcal{N}$
  - $\tilde{\mu}_R^{\mathcal{N}} = \int_{-\infty}^{t} \lambda e^{-\lambda(t-u)} \tilde{\mu}_R^{(u)} du$, where u is the specific cohort's time of entry.
- Applying the Stochastic Leibniz Rule, $\tilde{\mu}_R^{\mathcal{N}}$ perfectly resolves into an Ornstein-Uhlenbeck process:

$$d\tilde{\mu}_R^{\mathcal{N}} = (\lambda + \kappa)\left(\mu_R + \frac{\lambda}{\lambda + \kappa} s - \tilde{\mu}_R^{\mathcal{N}}\right) dt + \kappa \Sigma_R dB_X$$

- $\tilde{\mu}_R^{\mathcal{N}}$ is not "representative" when $\kappa > 0$: *wealth*-weighted aggregation is needed for market clearing
  - Possible Workarounds
    1. **Dynastic Wealth Pooling**. A "large family" structure that perfectly insures against birth-timing risk
    2. **Non-Conventional Demand**. Bypassing HJB, mechanically decoupling asset demand from wealth paths

Confirmation Bias $\kappa = 0$

- When $\kappa = 0$ (∃confirmation bias), type $\mathcal{N}$ investors share the same subjective expectation

$$\tilde{\mu}_R^{(i)} = \tilde{\mu}_R^{\mathcal{N}} = \boxed{\mu_R + s}$$

- Two Specifications for the prevailing subjective wedge $s(t)$
  1. Shared belief (exogenous. news, sentiment) driven by B_s with $dB_s \perp dB_f$. (DeLong et al., 1990)

$$ds = -\psi_n s dt + \sigma_s dB_s$$

  2. Market memory (exponentially weighted moving average of past returns. Malmendier&Nagel, 2016)

$$s(t) = \int_{-\infty}^{t} \psi e^{-\psi_m (t-u)} \left(dR(u) - \mu_R du\right)$$

  - Collapses exactly into an OU process via Leibniz: $ds = -\psi_m s dt + \psi_m \Sigma_R dB_X$

Wealth Dynamics and Optimization

- **Physical Wealth.** Individual i accumulates financial wealth under the true DGP:

$$dW^{(i)} = \left[(r+\lambda)W^{(i)} + \theta^{(i)}(\mu_R - r) - C^{(i)}\right] dt + \theta^{(i)}\Sigma_R dB_X$$

where $\theta^{(i)}$ is the dollar position in the risky asset, $C^{(i)}$ is consumption.

- **Planned Wealth.** When optimizing, individuals use their subjective expected return:

$$dW^{(i)} = \left[(r+\lambda)W^{(i)} + \theta^{(i)}(\tilde{\mu}_R^{(i)} - r) - C^{(i)}\right] dt + \theta^{(i)}\Sigma_R d\tilde{B}_X^{(i)}$$

(*Note.* $\tilde{\mu}_R^{(i)} = \mu_R$, $d\tilde{B}_X^{(i)} = dB_X$ for all type $\mathcal{R}$ agents)

  - **The Hamilton-Jacobi-Bellman Equation.** Agents maximize EU over *planned* wealth

$$(\rho+\lambda)V^{(i)} = \max_{\{C^{(i)},\theta^{(i)}\}} \left\{ \frac{(C^{(i)})^{1-\gamma}}{1-\gamma} + \frac{1}{dt}\mathbb{E}_t^{(i)}\left[dV^{(i)}\right] \right\}$$

Optimal Policies

- Factoring out $W^{(i)}$ so that $V^{(i)}(W,X) = \frac{(W^{(i)})^{1-\gamma}}{1-\gamma} v^{(i)}(X)$,

Optimal Consumption. $C^{(i)*} = (v^{(i)})^{-\frac{1}{\gamma}} W^{(i)}$

Optimal Risky Asset Position. $\theta^{(i)*} = \frac{W^{(i)}}{\gamma}\left(\frac{\tilde{\mu}_R^{(i)} - r}{\|\Sigma_R\|^2} + \frac{\Sigma_R \Sigma_X^\top \left(\nabla_X \ln v^{(i)}\right)}{\|\Sigma_R\|^2}\right)$

- Representative policies?

$$\begin{cases} C^{(i)*} = C^{\mathcal{R}}, & \theta^{(i)*} = \theta^{\mathcal{R}} \quad \forall i \text{ if } i \text{ is type } \mathcal{R} \\ C^{(i)*} = C^{\mathcal{N}}, & \theta^{(i)*} = \theta^{\mathcal{N}} \quad \forall i \text{ if } i \text{ is type } \mathcal{N} \textbf{ and } \tilde{\mu}_R^{(i)} = \tilde{\mu}_R^{\mathcal{N}} \text{ (representative belief)} \end{cases}$$

Aggregation and Demand

- Let $W^k = \int_{i \in k} W^{(i)} di$ for $k \in \{\mathcal{N}, \mathcal{R}\}$.
  - Assuming **representative policies** within type,

$$\int_{i \in k} C^{(i)} di = C^k = W^k c^k \quad \left(\text{where } c^k \equiv \frac{C^k}{W^k}, \text{ consumption-wealth ratio}\right)$$

$$\int_{i \in k} \theta^{(i)} di = \Theta^k = W^k \pi^k \quad \left(\text{where } \pi^k \equiv \frac{\Theta^k}{W^k}, \text{ fraction of wealth invested in the risky asset}\right)$$

- Define $\omega \equiv \frac{W^{\mathcal{N}}}{W^{\mathcal{N}} + W^{\mathcal{R}}}$. Risky asset market clearing implies (with the risk-free asset in zero net supply)

$$\omega \pi^{\mathcal{N}} + (1 - \omega) \pi^{\mathcal{R}} = 1$$

  - This condition holds universally, regardless of how the production economy is specified.

Goods Market Clearing

- Goods market clearing implies $C^{\mathcal{N}} + C^{\mathcal{R}} = (1-\tau)Y$.
  - Dividing both sides by aggregate wealth $W^{\mathcal{N}} + W^{\mathcal{R}}$ yields

$$\omega c^{\mathcal{N}} + (1-\omega)c^{\mathcal{R}} = \begin{cases} \frac{1-\tau}{p} & \text{(endowment)} \\ \frac{1}{q}(\bar{A}(1-\tau) - \iota - \frac{\phi_I}{2}(\iota-\delta)^2 & \text{(production)} \end{cases}$$

where $p \equiv \frac{P}{D}$ (price-to-output ratio) and $q \equiv \frac{P}{K}$ (Tobin's q) and $\iota \equiv \frac{I}{K}$.

Wealth Share Dynamics

- Applying Itô's Lemma to $\omega \equiv \frac{W^{\mathcal{N}}}{W^{\mathcal{N}}+W^{\mathcal{R}}}$ yields $d\omega = \mu_\omega dt + \Sigma_\omega dB_X$.
- The drift of wealth share (μ_ω)

$$\mu_\omega = \omega(1-\omega)\left[\left(\tilde{\mu}_W^{\mathcal{N}} - \mu_W^{\mathcal{R}}\right) - \Gamma\right] + \frac{\tau}{p}(\alpha - \omega)$$

  - $\Gamma = (1-\omega)(\pi^{\mathcal{N}} - \pi^{\mathcal{R}})\Sigma_R \Sigma_W^\top$ is the Itô variance-covariance correction term
  - The difference in expected wealth growth

$$\tilde{\mu}_W^{\mathcal{N}} - \mu_W^{\mathcal{R}} = (\pi^{\mathcal{N}} - \pi^{\mathcal{R}})(\mu_R - r) - (c^{\mathcal{N}} - c^{\mathcal{R}})$$

- The diffusion matrix of wealth share (Σ_ω)

$$\Sigma_\omega = \omega(1-\omega)(\pi^{\mathcal{N}} - \pi^{\mathcal{R}})\Sigma_R$$

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